Term Project — Mystery Mini-Organism Annotation

Gebze Technical University, Faculty of Engineering BENG451 — Introduction to Bioinformatics (Undergraduate) BSB511 — Bioinformatics Fundamentals (Graduate) Spring Semester 2026

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1. The setup

Your team works at a fictional bioinformatics consultancy. A client has dropped off a sample with an interesting backstory and a FASTA file of about twelve predicted protein-coding genes from its already-assembled genome. They want to know: **what organism is this, and what does it do?**

Each team has been given a different case file. Inside your team's folder you will find:

- `case_file.md` — a short brief from the client describing where the sample came from and what they want answered
- `mystery_orfs.fasta` — twelve unannotated open reading frames (protein sequences). The headers are opaque IDs (`mORF_01`, `mORF_02`, …). You will not find taxon information in the FASTA itself
- `README.md` — pointer to this document

Your job is to identify the organism, annotate its ORFs, place it phylogenetically, and tell us — clearly and defensibly — what its protein content reveals about its lifestyle.

This is the kind of work a real consultancy bioinformatician does on a Monday morning: opaque inputs in, well-reasoned conclusions out. The case file is fictional, but the sample provenance, every protein sequence in your FASTA, and every fact behind every backstory has been verified against published literature. There is a real answer.

2. What you must produce

You will hand in a single zip per team containing the following files. The naming convention uses your team's number (e.g. `team_03`).

File	Format	Required
`team_NN_pipeline.ipynb`	Jupyter notebook	Yes — the full analysis pipeline that runs end-to-end on your `mystery_orfs.fasta`
`team_NN_annotation_table.csv`	CSV	Yes — one row per ORF
`team_NN_marker_tree.nwk`	Newick	

File	Format	Required
		Yes — the phylogenetic tree used to identify the organism
`team_NN_report.pdf`	PDF	Yes — 4–5 pages of body text (excluding figures and references)
`team_NN_slides.pptx`	PowerPoint	Yes — the slides for your in-class talk

In addition, every team will give a **10-minute presentation followed by a short Q&A** during a presentation session announced separately.

The deliverables together form one piece of work: the notebook is the evidence, the table summarizes it, the tree carries the identification claim, the report is the scientific case, and the talk is how you defend it in front of others.

3. The required pipeline

You must produce a single Jupyter notebook that runs end-to-end on your `mystery_orfs.fasta` file. The notebook has six steps below. The project asks you to integrate the techniques you have learned this semester into a single coherent workflow.

Step 1 — Load and inspect

Read the FASTA with `Bio.SeqIO`. Print a sanity table showing each ORF's ID, length, and first thirty residues. This is the trivial sanity-check that confirms your data parsed correctly.

Step 2 — Per-ORF homology search

The project is **web-API based** — you do not need to install BLAST+, HMMER, or any large local database. Both required tools below run on public servers.

For every ORF, perform two searches:

1. **A protein-against-protein BLAST search (`blastp`) via the NCBI web service.** Use `Bio.Blast.NCBIWWW.qblast` with `program="blastp"` and `database="swissprot"`. Parse the output with `Bio.Blast.NCBIXML.read`. From each BLAST result extract: the top three hits (organism + description), the best E-value, the best percent identity, and the best query coverage.
2. **A Pfam-A domain scan via the EBI InterProScan REST API.** Submit the sequence with `appl=PfamA`, poll the job until it finishes (status endpoint), and fetch the TSV result. Extract the matching Pfam family name(s) and accession(s), their E-values, and the domain coordinates. The submit-poll-fetch pattern is shown in the example code below.

Record one row per ORF in a pandas DataFrame.

Politeness rules — these are not optional. All public web APIs throttle clients that abuse them; please follow these rules so your team's IP address (and your University's) is not blocked:

- Set `Bio.Entrez.email = "your@gtu.edu.tr"` (your real email) at the top of your notebook. NCBI requires this and will refuse anonymous bulk requests.
- Pass `email="your@gtu.edu.tr"` in every InterProScan submission.
- Do not run more than **three** search jobs in parallel.
- Sleep at least **3 seconds** between submissions. Poll job status no faster than **once every 5 seconds**.
- **Cache** every result to disk after the first run (e.g. write each `qblast` XML and each InterProScan TSV to a per-ORF file). Re-running your notebook should NOT re-submit jobs that have already completed.

A simple way to follow the cache rule:

```
def cached(out_path, fetch_fn):
    if Path(out_path).exists():
        return Path(out_path).read_text()
    text = fetch_fn()
    Path(out_path).parent.mkdir(parents=True, exist_ok=True)
    Path(out_path).write_text(text)
    return text
```

Wrap each web call with `cached(...)`.

Step 3 — Twilight-zone follow-up

For ORFs whose best `blastp` E-value is greater than 1e-3, **or** whose top BLAST hit description is uninformative ("hypothetical protein", "uncharacterized protein", "DUF…"), run **PSI-BLAST** as an iterative follow-up. PSI-BLAST is available through the same NCBI web service: pass `service="psi"` and `word_size=3` to `Bio.Blast.NCBIWWW.qblast`. (The web service handles up to ~3 PSI-BLAST iterations server-side; if you want explicit iteration control, you may instead drive the NCBI BLAST web URL directly.) Record whether PSI-BLAST recovered a clearly-named family that simple BLAST missed.

This is the step that handles cases where simple BLAST alone is not enough — the kind of case you saw in lecture under the heading "twilight zone" (around 20% identity, where homology becomes hard to detect with pairwise alignment alone). PSI-BLAST is the iterative-profile method we learned in lecture; the underlying idea is identical to the iterative HMM-profile approach (`jackhmmer`).

(If you would prefer to use `jackhmmer` via the EBI HMMER REST API instead of NCBI PSI-BLAST, that is also acceptable; document your choice. Both are iterative profile methods. Do not run them locally — use the web service.)

Step 4 — Annotation table

Save your per-ORF findings to disk as `team_NN_annotation_table.csv` (or Markdown). One row per ORF. Required columns at minimum:

```
orf_id, length, blast_top_hit, blast_evalue, blast_pct_id,
pfam_hits, pfam_evalue, twilight_followup_used, twilight_followup_outcome,
proposed_annotation, evidence_summary, confidence
```

`confidence` is a three-level human label (`high`, `medium`, `low`) — your judgment based on the evidence. There is no formula. An ORF with a clear BLAST hit at E < 1e-50 to a single well-named protein, plus matching Pfam domains, is `high`. An ORF that only converged through PSI-BLAST onto a poorly-characterized family is `low`. Most cases will sit in between.

Step 5 — Phylogenetic placement

Pick **one** of your ORFs that gave a clean BLAST signal (typically a housekeeping gene like RpoB, DnaK, EF-Tu, or a ribosomal protein). This is your "marker gene". Then:

1. Use the NCBI BLAST web service (or a manual UniProt search) to gather ten to fifteen orthologs of your marker across taxa, deliberately spanning the candidate clade implied by the case file. If your case file mentions a hot environment, your ortholog set should include several thermophiles, an archaeon, and at least one mesophile outgroup.
2. Build a multiple sequence alignment. The simplest web-based option is the **EBI MUSCLE REST API** at `https://www.ebi.ac.uk/Tools/services/rest/muscle/run/` (submit-poll-fetch, the same pattern as InterProScan). The same submit/poll/fetch helper code from Step 2 will work — just point it at the MUSCLE endpoint. If you have MUSCLE installed locally already, that is also fine.
3. Compute a distance tree (UPGMA or Neighbor-Joining) using `Bio.Phylo.TreeConstruction.DistanceCalculator` followed by `DistanceTreeConstructor`. (This is local Biopython, not a web call — no API needed.)
4. Render the tree with `Bio.Phylo.draw` and export it as Newick (`team_NN_marker_tree.nwk`).
5. **State your team's identification.** A clean form: "Based on the marker tree, we believe this sample is from Genus species (or Genus sp.)." Be honest: if your tree only resolves the genus and not the species, say so.

Step 6 — Interpretation

Inside the notebook (as Markdown cells, expanded later in your PDF report), address all five of the following:

1. **Identification call.** What organism, and what is the strongest single piece of evidence?

2. **Lifestyle inference.** What does your ORF set tell you about the organism's environment, metabolism, or biology? Cite specific ORFs.
3. **Spotlight finding.** Pick one ORF that you find biologically interesting given the inferred lifestyle, and explain why it is interesting. ("Why is this protein in this organism?")
4. **Method comparison.** Which ORFs did BLAST alone solve cleanly? For which ORFs did Pfam or PSI-BLAST add the decisive evidence? Quote one specific ORF as an example.
5. **Limits and confidence.** Which ORFs remain unidentified or ambiguous, and what additional evidence or methods would resolve them?

4. Allowed and disallowed

Allowed and recommended (web-based). This is the path the project is designed around:

- **Biopython** — `SeqIO`, `Entrez`, `Blast.NCBIWWW.qblast` (with `program="blastp"` and with `service="psi"` for PSI-BLAST), `Blast.NCBIXML`, `AlignIO`, `Phylo`, `Phylo.TreeConstruction`.
- **EBI InterProScan REST** for the Pfam scan (`appl=PfamA`).
- **EBI MUSCLE REST** for the marker MSA (or any equivalent web aligner).
- Standard scientific Python: `pandas`, `numpy`, `matplotlib`, `requests`.

Politeness with the public web APIs is mandatory — see Step 2 above. Set your real email; cap concurrency at 3; cache every result; sleep between submissions.

Allowed but not required. ScanProsite (`Bio.ExPASy.ScanProsite`) for motif analysis if you find it useful. Local BLAST+, MUSCLE, HMMER, etc. only if you happen to have them installed already — but the project does **not** assume any local installation.

AI assistance. AI tools (ChatGPT, Claude, Copilot, etc.) are permitted for coding assistance. This is consistent with the AI policy you saw in Lecture 1 and in Homework 1. **You remain responsible for the correctness of every claim and every line of code in your submission.** If your AI assistant hallucinates a protein name, an organism, or a result, and you do not catch it, the points are lost. Verify everything against the actual tool output.

Disallowed. Structure prediction (AlphaFold, PyMol, I-TASSER, SWISS-MODEL) is out of scope for this semester — we will reach this in weeks 11–13. You may mention a structural argument, but you may not build a part of your project on a predicted structure. Also: do **not** submit hundreds of unthrottled requests to NCBI or EBI — submissions found to abuse the public APIs (no email, no caching, no delays) will lose pipeline marks.

5. Recommended report structure

Your PDF report should be 4–5 pages of body text (figures and references in addition). The expected sections, with rough page budgets:

1. **Case file and question** (~½ page) — restate the brief and frame the question
2. **Methods** (~1 page) — the steps of your pipeline, the tools and versions used, the parameters chosen and why
3. **Annotation results** (~1 page) — the summary table plus one or two figures (a domain-architecture plot is a good choice)
4. **Phylogenetic placement and identification** (~1 page) — the tree, what it shows, and your identification claim
5. **Discussion** (~1–1½ pages) — lifestyle inference, the spotlight finding, your method comparison, and the limits of confidence

This structure is recommended, not enforced — but if you deviate, every section above must still appear somewhere in your report.

6. Recommended presentation structure

A 10-minute talk (then Q&A) should hit, in order:

1. The case file — where did the sample come from and what was the question?
2. Pipeline overview — the six steps, briefly
3. Headline result — your organism identification
4. Evidence — the marker tree plus one or two illustrative ORFs
5. Spotlight finding — your one biologically interesting ORF
6. Limits — what you could not resolve, and what would help

Practice your timing. Ten minutes goes faster than you think.

7. Grading

Five weighted components, totalling 100%.

Component	Weight	Full-marks criterion
Pipeline notebook	25%	Runs end-to-end without manual intervention; well-organized; correct use of Biopython, NCBI BLAST, and EBI InterProScan; **proper API politeness** (email set, results cached, no abusive parallelism); reasonable parameter choices documented as

Component	Weight	Full-marks criterion
		comments or markdown cells
Annotation table	20%	All ORFs annotated; required columns populated; PSI-BLAST (or equivalent) used appropriately on twilight-zone ORFs; confidence labels are defensible given the evidence
Phylogeny + organism identification	20%	Sensible marker gene; orthologs chosen to span the candidate clade; tree built correctly; identification supported by tree topology, not asserted
Report (PDF)	20%	Clear scientific writing; figures earn their space; lifestyle inference and spotlight finding are well-argued; method comparison cites specific ORFs
Presentation	15%	Clear; well-budgeted in time; visuals are legible; Q&A demonstrates that you own the work

The rubric measures defensibility more than correctness. An identification that is half-right but well-reasoned with honest discussion of uncertainty will score above an identification that is fully right but presented as a foregone conclusion.

8. A closing thought

The whole point of this project is that none of you have done it before. You may be wrong. Your tree may not resolve cleanly. Your ORFs may not all annotate. Your spotlight finding may not be as exciting as you hoped. **That is fine.** What we want to see is the reasoning — how you used the tools you learned, how you made judgment calls under uncertainty, how you communicated what you concluded and what you could not.

Have fun with the case file. The mystery is real. The answer is real. We look forward to seeing what you make of it.

— Course staff, BENG451 / BSB511